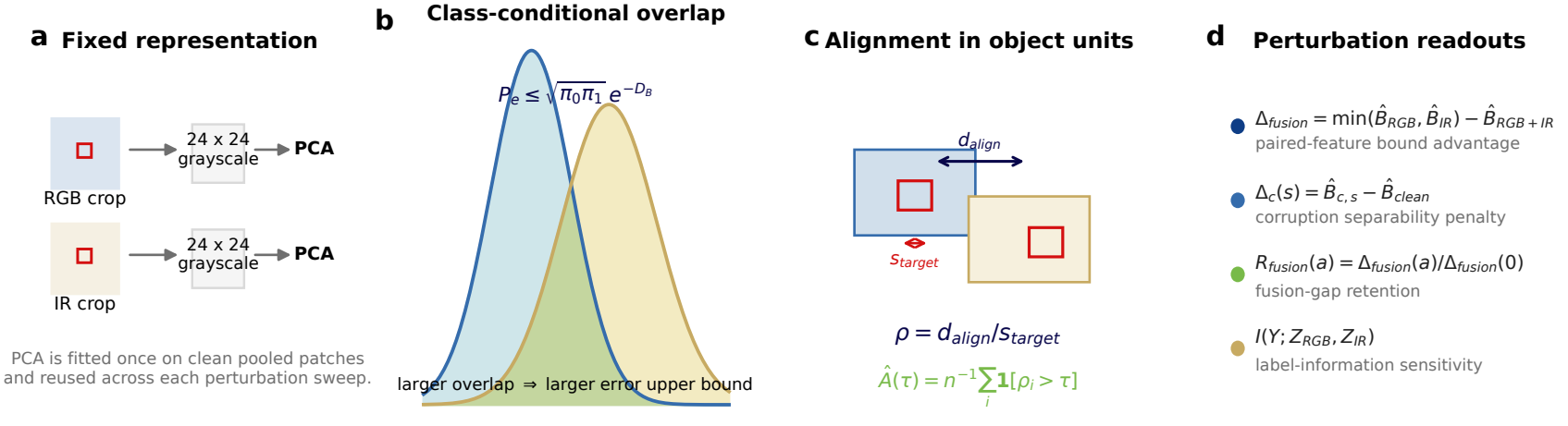
a Fixed representation
RGB crop
24 x 24 grayscale
PCA
IR crop
24 x 24 grayscale
PCA
PCA is fitted once on clean pooled patches and reused across each perturbation sweep.
b Class-conditional overlap
$P_e \le \sqrt{\pi_0 \pi_1}\, e^{-D_B}$
larger overlap ⇒ larger error upper bound
c Alignment in object units
$d_{align}$
$s_{target}$
$\rho = d_{align}/s_{target}$
$\hat{A}(\tau) = n^{-1} \sum_i \mathbf{1}[\rho_i > \tau]$
d Perturbation readouts
$\Delta_{fusion} = \min(\hat{B}_{RGB}, \hat{B}_{IR}) - \hat{B}_{RGB+IR}$
paired-feature bound advantage
$\Delta_c(s) = \hat{B}_{c,s} - \hat{B}_{clean}$
corruption separability penalty
$R_{fusion}(a) = \Delta_{fusion}(a)/\Delta_{fusion}(0)$
fusion-gap retention
$I(Y; Z_{RGB}, Z_{IR})$
label-information sensitivity